

Action Recognition and Benchmark Using Event Cameras

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Abstract—Recent years have witnessed remarkable achievements in video-based action recognition. Apart from traditional frame-based cameras, event cameras are bio-inspired vision sensors that only record pixel-wise brightness changes rather than the brightness value. However, little effort has been made in event-based action recognition, and large-scale public datasets are also nearly unavailable. In this paper, we propose an event-based action recognition framework called EV-ACT. The Learnable Multi-Fused Representation (LMFR) is first proposed to integrate multiple event information in a learnable manner. The LMFR with dual temporal granularity is fed into the event-based slow-fast network for the fusion of appearance and motion features. A spatial-temporal attention mechanism is introduced to further enhance the learning capability of action recognition. To prompt research in this direction, we have collected the largest event-based action recognition benchmark named THU^{E-ACT}-50 and the accompanying THU^{E-ACT}-50-CHL dataset under challenging environments, including a total of over 12,830 recordings from 50 action categories, which is over 4 times the size of the previous largest dataset. Experimental results show that our proposed framework could achieve improvements of over 14.5%, 7.6%, 11.2%, and 7.4% compared to previous works on four benchmarks. We have also deployed our proposed EV-ACT framework on a mobile platform to validate its practicality and efficiency.

Index Terms—Action recognition, dynamic vision sensor, event camera, event representation.

I. INTRODUCTION

ACTION recognition is an active research topic with a wide range of applications in visual surveillance, video analysis and retrieval, healthcare, and abnormal activity recognition [1],

[2], [3], [4] in the past decades. A large number of well-performed frame-based action recognition methods have been proposed, achieving increasingly better performance on many well-known benchmarks. However, considering the efficiency and power consumption, it is not necessary for sensors and intelligent systems to analyze every frame of the video due to the rarity of anomalous and noteworthy behaviors. Due to the nature of RGB images, it is difficult to determine whether there is a noticeable action in a specific image, which forces us to save and feed every moment of the image into deep networks for real-time applications. Moreover, the proliferation of surveillance sensors and devices in both indoor and outdoor scenarios has raised concerns about the violation of users' privacy by automated action analysis systems, which use frame-based images that directly expose users' appearance and identity. These concerns make deploying frame-based action recognition in real-world applications challenging.

As an alternative to traditional frame-based cameras, bio-inspired event cameras, e.g., Dynamic Vision Sensors (DVS) [5], [6], have created a new visual paradigm. Different from frame-based cameras that capture images at a fixed exposure rate, event cameras respond to the pixel-wise brightness changes asynchronously, and output in the form of a tuple list, called *event stream*. Event cameras only record the moving part of the scene while the invariant background is discarded, therefore the storage and computational load are considerably reduced. Moreover, since the event streams do not contain any color and texture features that directly reveal personal appearance, it is considered beneficial in certain scenarios where privacy is valued, e.g., detecting falling down for the elderly in bathrooms, which can be informed in Fig. 1. To some extent, event cameras and event-based action recognition methods are deemed even more suitable for practical deployments than traditional frame-based cameras and action recognition methods.

Although various deep learning methods have been proposed for frame-based action recognition, they cannot be directly used in event-based action recognition. The first challenge lies in the lack of datasets. Since event cameras have only emerged in the commercial market in the last decade, event-based datasets are less available in various computer vision fields. While PAF [7], DHP19 [8], N-HAR [9], and DailyAction [10] are four publicly available event-based datasets that can be applied to action recognition, both categories and data in these datasets are insufficient, and the action categories are relatively basic and not

Manuscript received 11 September 2022; revised 25 May 2023; accepted 27 July 2023. Date of publication 1 August 2023; date of current version 3 November 2023. This work was supported by the National Key Research and Development Program of China under Grant 2020AAA0108100. Recommended for acceptance by C. Fermüller. (Corresponding author: Shaoyi Du.)

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Digital Object Identifier 10.1109/TPAMI.2023.3300741

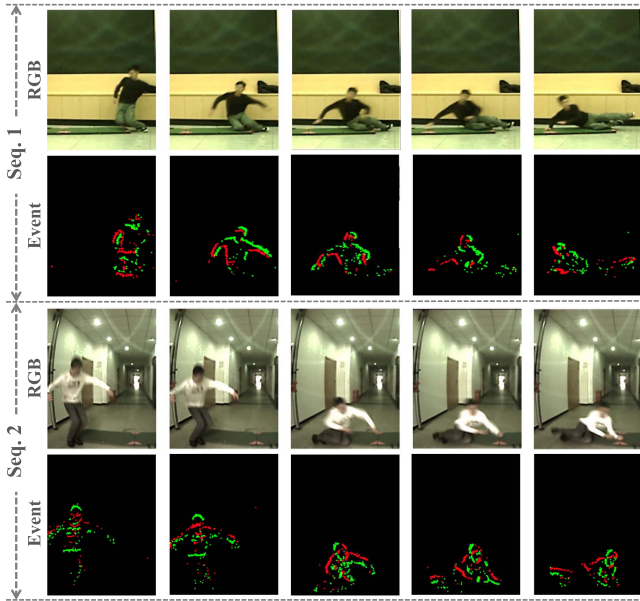


Fig. 1. Examples of “falling down” in two scenarios from the released THU-E-ACT-50 benchmark. The event camera utilizes only two polarities (red and green) to represent the person’s motion, with the potential for low power consumption and privacy encryption.

event-specific. Additionally, event data often holds redundant noise, leading to excessive burdens on the network. Another challenge is the lack of effective representation and action recognition frameworks designed for event streams. As the sparse and asynchronous event camera is fundamentally different from the dense and synchronous frame-based cameras, it is impossible to feed the event stream directly into CNNs. Several works [11], [12], [13], [14], [15] proposed to convert event streams into grid-based event frames, but these are handcrafted simple features, and there is no consensus on which representation is most effective to use.

In this paper, we contribute to both the development of a new large-scale dataset and the event-based action recognition framework. Although there are some existing event-based action recognition datasets, they are either small (less than 3,000 samples per dataset) or with few action types (5 to 33 categories). Therefore, a large benchmark for event-based action recognition is urgently required. We have created two large-scale real-world event-specific action recognition benchmarks, namely THU-E-ACT-50 and THU-E-ACT-50-CHL, which both contain 50 action categories. The standard THU-E-ACT-50 covers 10,500 captured sequences at a resolution of 1280×800 , while the THU-E-ACT-50-CHL comprises 2,330 sequences at a resolution of 346×260 recorded in challenging conditions. Several actions for healthcare monitoring are specially included, e.g., falling down, stomach pain, headache, nausea, back pain, and staggering. In addition to single-player actions, two-player interactive actions are also integrated, including handshakes, fighting, handing over objects, lifting chairs, etc. We have also proposed a new event representation, called Learnable Multi-Fused Representation (LMFR), and an action recognition framework using event cameras, called EV-ACT. Our proposed

framework consists of four modules, including an event voxel filter module, a learnable multi-representation fusion module, an event-based slow-fast network, and an event-based spatial temporal attention mechanism, capable of learning the multi-fused representation of the event stream and performing action classification. Experiments on four benchmarks, i.e., PAF [7], DHP19 [8], DailyAction [10] and our large-scale THU-E-ACT-50 demonstrate the superiority of the proposed framework over the state-of-the-art, achieving improvements of more than 14.5%, 7.6%, 11.2%, and 7.4%, respectively. Moreover, we also demonstrate the practicality of EV-ACT by deploying it on a mobile platform. The constructed benchmarks can be accessed at.¹

Overall, the main contributions of this paper can be summarized as follows:

- We propose a CNN-based event-based action recognition framework that integrates diverse event representations in a learnable manner, satisfying the need for low power consumption without exposing fine-grained visual details. An event-based slow-fast network and event spatial temporal attention mechanism are further proposed to effectively extract and fuse multi-granularity features.
- We contribute the largest event-based action recognition benchmark and an ancillary dataset for public research use, containing over 12,830 recordings captured under diverse illuminations.
- We demonstrate the proposed method’s capability for real-time, low-power action recognition in practical applications by implementing an embedded system.

II. RELATED WORK

A. Action Recognition

The pipeline for frame-based action recognition can be decomposed into two steps, i.e. action representation and action classification [16]. Among earlier approaches, the former step converts the video into a feature vector [17], [18], [19], [20], while the latter infers a label from the vector [21], [22]. Recently, deep networks have combined these two modules to form the end-to-end learning framework, dramatically improving the performance of action classification. However, the main challenge in video action recognition is how to model temporal correlations between frames due to the inability to utilize information from all frames. 3D CNNs [23] learn features in both the spatial and temporal domains but are computationally expensive. SlowFast [24] achieves performance enhancement by setting fast and slow pathways to fuse features on different sampled rates. TSM [25] proposes to shift part of the channels along the temporal dimension, which can achieve a better compromise between performance and energy consumption. Nevertheless, these mainstream action recognition networks still require continuous high-power processing, which poses challenges for practical applications.

B. Event Representation

Since the event stream is in the form of a tuple list, its processing is fundamentally different from traditional images.

¹<https://gaoyue.org/dataset/THU-EACT-50>

One option is to use Spiking Neural Networks (SNNs) [26], [27], [28] since they are well-suited to process spike trains. However, SNNs require special hardware for deployment and are harder to train than CNNs. Another approach is to analyze the timestamp and polarity of events for representation, including local plane fitting [12] and temporal derivative interpolation [29], but these methods are task-specific and not easily generalized to other applications. The most widely used approach is to convert the raw event data to 2D or 3D grid-based representations suitable for deep network input. The first method is called event counting [14], [30], i.e., counting the number of events in each polarity to form two $H \times W$ frames, which preserves the polarity but loses temporal information. Similarly, event frames [11] are formed by summing the polarities of events within a specific time window. The voxel grid-based approach uses a bilinear sampling kernel to map events to the nearest temporal grid [15]. Another commonly used type of representation is called the time surface [31], [32], which encodes timestamps as pixel values. The Surface of Active Events (SAE) [12] records the time of the latest event at each pixel location. The memory surface [9] uses three different types of time-based kernels (binning, linear, and exponential) that can be used across both time and event indexes. The time image [33] is another representation that depends not only on events but also on the motion hypothesis. Further, [34] proposes TORE volumes that record the timestamp of the latest three events. Typically, these methods rely on quantization or windowing, where fixed intervals are set in the temporal domain, and the asynchronous event stream is accumulated into frame-like representations. However, most of these methods are based on hand-crafted features and there is currently no definitive conclusion as to which one is the best because of the different information captured.

C. Datasets for Action Recognition

Datasets are important contributions to driving progress in deep learning methodologies. For frame-based action recognition, there are many existing well-established datasets. KTH [36] is an early action dataset containing videos of 6 action categories at 160×120 resolution in different scenes. I3DPost [37] provides videos of two people interacting with each other for 8 actions. UCF50 and UCF101 [1] contain 50 and 101 action categories from YouTube, respectively. Kinetics [3] consists of a series of large-scale datasets released by DeepMind, containing 400 action categories with over 400 videos per action. Although datasets of conventional image-like videos are abundant, the currently available action recognition datasets based on event data are inadequate. PAF [7] is the first action recognition dataset captured by an event camera, with 10 actions and a total of 450 recordings. DHP19 [8] is an event dataset oriented to the pose estimation task, which contains 33 sub-actions and a total of 557 videos. Although it can be used for the action recognition task, its categories are less applicable in practice and the data from certain viewpoints (e.g., backside) are not well suited for action recognition. N-HAR [9] is another category-unbalanced indoor dataset with 3,091 videos, but containing only 5 actions. Recently, DailyAction [10] has been released for event-based

action recognition having 12 actions and 1,440 recordings, but at a low resolution of only 128×128 . A large-scale event-based action recognition dataset in terms of action categories and recording samples is essential to drive research in this area.

III. METHOD

To utilize the raw event data for action recognition, we propose a CNN-based framework for event-based action recognition, including the Event Voxel Filtering module, the Learnable Multi-Representation Fusion module, the Event-based Slow-Fast Network, and the Event-based Spatial Temporal Attention mechanism, as illustrated in Fig. 2. The Event Voxel Filtering module implements event denoising and substantial down-sampling while preserving the event distribution. The proposed Learnable Multi-Fusion Representation (LMFR) integrates multiple mainstream event representations in a learnable manner, transforming raw event data into a frame-like input format for CNNs. The Event-based Slow-Fast Network exploits fast and slow features in human motion and appearance. The Event-based Spatial Temporal Attention mechanism is introduced to further improve performance.

A. Event Cameras and Event Stream

Event cameras are bio-inspired vision sensors that respond to brightness changes at each pixel asynchronously. For the pixel (x, y) at timestamp t , the event camera triggers an event when

$$L(x, y, t) - L(x, y, t') > p \cdot \theta, \quad (1)$$

where $L(x, y, t)$ denotes the log brightness, t' is the time of the last event triggered at (x, y) , θ is a constant threshold, $p \in \{-1, +1\}$ represents the polarity of the event, indicating the brightening is increasing or decreasing. Each event is recorded as a tuple $e = (x_k, y_k, t_k, p_k)$, and the event stream can be represented as a tuple list E :

$$E = \{e_k\}_{k=1}^N = \{(x_k, y_k, t_k, p_k)\}_{k=1}^N. \quad (2)$$

The event camera does not capture any color or texture information, thus providing a privacy advantage compared to traditional frame-based cameras. Meanwhile, since the event camera only outputs events when the intensity changes, few events are triggered when no valid movement. It is estimated that event cameras consume approximately 150 times less power than standard cameras and have significantly lower bandwidth requirements [38]. Furthermore, in some high-speed scenes, e.g., when someone drops something, the event cameras can capture it very well without motion blur like RGB cameras due to their working principle. All these advantages allow the event camera to be used as a low power-consumption, high-efficiency, and privacy-encrypted action recognition vision sensor.

B. Event Voxel Filtering

Compared with frame-based video action recognition methods that require sampling to obtain key frames, it is feasible to process all events within a time period since the event stream does not have redundant data that remain unchanged. However,

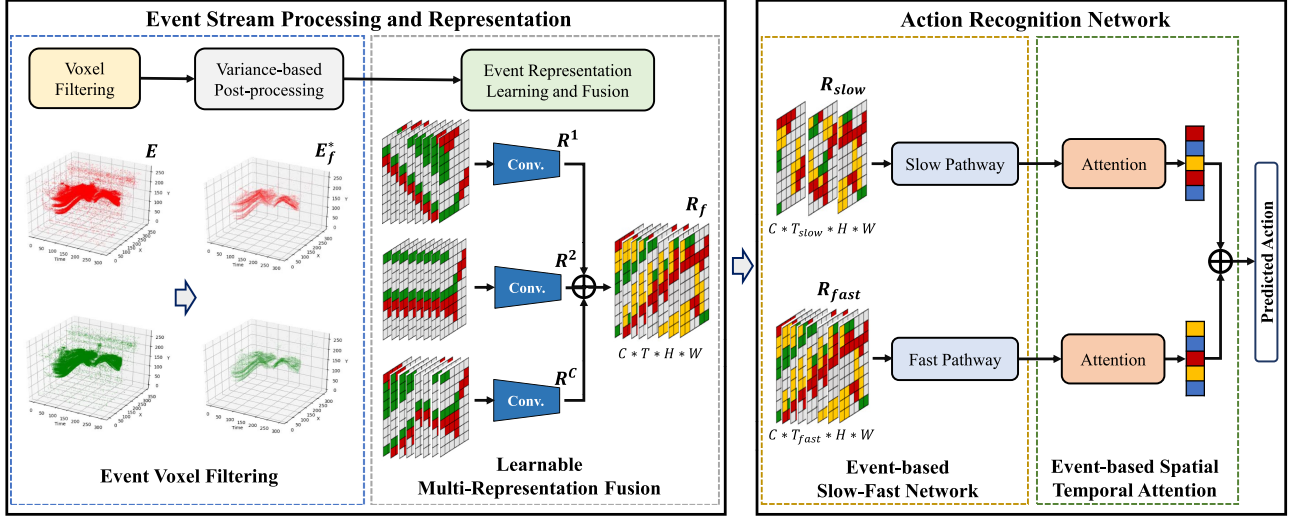


Fig. 2. Pipeline of the proposed EV-ACT, including an event voxel filtering module, a learnable multi-representation fusion module, an event-based slow-fast network, and an event-based spatial temporal attention mechanism.

the current event cameras are noisy, which affects the accuracy of recognition. To address this issue, the method proposed in [39] involves down-sampling by retaining only the edges that are parallel to each event point on the time plane. Despite this, the method is based on GCN networks for edge elimination rather than point-specific and requires traversing all event points. Inspired by the idea of point clouds voxel filtering [40], we introduce an Event Voxel Filtering (EVF) module to both down-sample and denoise the event stream.

Specifically, our proposed EVF module treats each event (x_k, y_k, t_k, p_k) as a three-dimensional point (x_k, y_k, t_k) with polarity p_k as the attribute, so the event stream can be regarded as a particular kind of point cloud. The X-Y-T space is subdivided into $D_x \times D_y \times D_t$ voxels of varying sizes according to the local density of the event stream. The average coordinates value of all events within a voxel is used to represent all the events in that voxel. Mathematically, the result of our proposed EVF module can be formulated as:

$$E_f^p = \{e_i\}_{i=1}^{N'} = \{(x_i^p, y_i^p, t_i^p, p)\}_{i=1}^{N'}, \quad (3)$$

where $p \in \{+1, -1\}$ represents positive or negative respectively, E_f^p denotes the filtered event stream, N' is the number of voxels, x_i^p, y_i^p and t_i^p are formulated by:

$$\begin{cases} x_i^p = \frac{1}{N_i^p} \sum_{k=1}^{N_i^p} x_k \\ y_i^p = \frac{1}{N_i^p} \sum_{k=1}^{N_i^p} y_k \\ t_i^p = \frac{1}{N_i^p} \sum_{k=1}^{N_i^p} \alpha F(t_k) \end{cases}, \quad (4)$$

where $F(\cdot)$ denotes a normalization operator for timestamps, N_i^p is the count of events of polarity p in the i -th voxel, and α is a constant term adjusting the scale. There are two main differences between our proposed EVF module and the traditional point cloud voxel filtering. First, since the timestamp t has a different physical meaning from the spatial coordinates x and y , we regularize t and rescale it comparable to x and y .

Second, since events have two different polarities, we propose to handle them separately. To further diminish noise events, we leverage a variance-based post-processing strategy, which counts the number of events in each voxel and eliminates all points in those voxels whose number is less than 3 standard deviations.

C. Learnable Multi-Representation Fusion

Since the event stream is in the form of a tuple list, an event representation method must be designed to convert it to a grid-based tensor before feeding into the CNN. Among the approaches for event intermediate representation, most current approaches record different information by quantifying the event points. For example, the Event Frame [11] counts triggered events in each time window and constitutes an $H \times W$ representation. The Event Count [14], [30] counts event points by polarity, producing a $2 \times H \times W$ representation that retains polarity information. However, both methods lose the temporal information. In addition to the cumulative number of event points, the Surface of Active Events (SAE) [12] is a two-dimensional representation that uses event trigger timestamps as values and is considered to indirectly reflect motion trajectory. The Event Voxel [15] divides spatio-temporal space into $T \times H \times W$ three-dimensional voxels and maps events to the nearest voxel using bilinear sampling. As a further extension, Event Spike Tensor (EST) [35] preserves both time and polarity, producing a representation with dimension $2 \times T \times H \times W$. In terms of the evolution of intermediate representations, higher dimensions indicate more information retention, as can be inferred from Table I.

However, most of these representations are hand-crafted features, making it difficult to determine the optimal one due to variations in the recorded event information, e.g., polarity, event counts, and timestamps. Suppose the filtered event stream E_f obtained by our EVF module is divided into T time bins with

TABLE I

COMPARISONS OF VARIOUS EVENT INTERMEDIATE REPRESENTATIONS, WHERE H AND W REPRESENT THE HEIGHT AND WIDTH OF THE REPRESENTATION, T IS THE NUMBER OF TIME WINDOWS, AND C DENOTES THE NUMBER OF FUSED REPRESENTATIONS

Representation	Dimensions	Descriptions
Event Frame [11]	H*W	Record the event polarity on each pixel
Event Count [14], [30]	2*H*W	Record the event counts on each pixel for two polarities
Event Voxel [15]	T*H*W	Record the event counts via voxel grid summation
Surface of Active Events (SAE) [12]	2*H*W	Record the most recent event timestamp on each pixel for two polarities
Memory Surface [9]	2*H*W	Record the most recent event timestamp with three kernels on each pixel for two polarities
Time Image [33]	2*H*W	Record the motion-compensated event timestamp on each pixel for two polarities
Event Spike Tensor (EST) [35]	2*T*H*W	Record the most recent event timestamp via voxel grid summation for two polarities
LMFR (Ours)	C*T*H*W	Record event information by incorporating multiple representations

equal size Δt . For an event set $\mathcal{E}^*$ in a particular time bin, we summarize all events in it to form a pure representation. Actually, almost all current event intermediate representations can be generalized to the following expression.

$$R = \sum_{e_i \in \mathcal{E}^*} F_{\pm}(x, y, t) \cdot f(x_i - x, y_i - y, t_i - t) \quad (5)$$

where the first term $F_{\pm}(x, y, t)$ denotes a pure representation, such as $F_{\pm}(x, y, t) = \pm 1$ in the Event Frame [11], $F_{\pm}(x, y, t) = 1$ in the Event Count [14], [30], and $F_{\pm}(x, y, t) = (t - t_0)/\Delta t$ in the Surface of Active Events (SAE) [12]. The second term $f(x_i - x, y_i - y, t_i - t)$ is a kernel function that can transform the pure representation, which is adopted only in a few works, such as the alpha-kernel function [41], [42], denoted as $k(x, y, t) = \delta(x, y)(et/\tau)\exp(-t/\tau)$, and the exponential kernel function [32], [43] calculated by $k(x, y, t) = \delta(x, y)(1/\tau)\exp(-t/\tau)$.

In this paper, we propose a novel event representation called Learnable Multi-Fused Representation (LMFR) that can incorporate multiple basic event representations $[R^1, R^2, \dots, R^C]$ and has learning capabilities, as presented in Fig. 2. Specifically, each pure event representation is unified into $T \times H \times W$ dimension, and a learnable convolutional layer is used to implement the kernel function that better transforms each pure representation. Thus, the j th basic event representation R^j can be expressed as

$$R^j = \sum_{e_i \in \mathcal{E}^*} F_{\pm}^j(x, y, t) \cdot \text{Conv}^j(x_i - x, y_i - y, t_i - t) \quad (6)$$

where $\text{Conv}^j(x_i - x, y_i - y, t_i - t)$ is the convolution kernel for the j th basic event representation. The final event representation R_f is represented as a concatenation of all transformed basic event representations along the channel dimension, denoted as:

$$R_f = [\phi(R^1); \phi(R^2); \dots; \phi(R^C)] \quad (7)$$

where $\phi(\cdot)$ is the batch normalization layer for normalization. It can be inferred that the proposed LMFR representation records event information with dimensions of $C \times T \times H \times W$ by integrating diverse mainstream representations, which can be easily fed into the CNN-based networks.

D. Event-Based Slow-Fast Network

Inspired by SlowFast [24], a classical structure in frame-based video action recognition, we propose the Event-based Slow-Fast Network (ESFN) for exploring feature augmentation with dual temporal granularities. Due to the sparsity of the event stream, different granularities of the length of the time bin Δt may yield varying representations. A larger Δt , i.e. a smaller number of time bins T , can accumulate more spatial semantic information, while a smaller Δt acutely captures fast movements in a short period of time. Similarly, the contours and motion trajectories of objects require a large time window to accumulate events and depict them, so the slow pathway focuses on the categorical spatial semantics of identity, e.g., recognizing “waving hands” requires constant monitoring of the trajectory of limb movement, which is difficult to realize from the event stream within a short time bin. On the other hand, some high-speed actions, e.g. “dropping an object” and “falling down”, need to be captured by the fast pathway, as they depend on the details of a very short time period.

In practice, event representations with a large time window Δt are sent to the slow pathway to extract semantic boundary information within the scene, while event representations with a small time window Δt are sent to the fast pathway to extract detailed high-speed action information. Mathematically, assume the learned event representation of slow pathway and fast pathway be R_{slow} and R_{fast} with dimension $C \times T_{slow} \times H \times W$ and $C \times T_{fast} \times H \times W$, respectively, while the backbone networks of the slow and fast branches are denoted as $M_{slow}(\cdot)$ and $M_{fast}(\cdot)$. Then, the extracted features derived from the backbone network without passing through the final pooling and fully connection layer are defined as:

$$R'_{slow} = M_{slow}(R_{slow}) \quad (8)$$

$$R'_{fast} = M_{fast}(R_{fast}) \quad (9)$$

where R'_{slow} is the extracted feature of slow pathway with dimension $C \times T_{slow} \times H' \times W'$, while R'_{fast} is the extracted feature of fast pathway with dimension $C \times T_{fast} \times H' \times W'$. The slow-fast dual-branch structure has been experimentally demonstrated to be effective in event-based action recognition as well.

E. Event-Based Spatial Temporal Attention

The attention mechanism has been extensively explored in various fields in recent years. Since the event camera only responds to moving objects, the attention mechanism will be more effective in sparse event data. When a person or target is moving, only the regions containing movement trigger a large number of events that require particular attention, while the rest areas containing a huge amount of background noise can be filtered out. While in the temporal dimension, since events exist only in the time bins with motion, it is unnecessary to focus on those inactive bins where nothing happens.

For these reasons, we propose a simple but effective Event-based Spatial Temporal Attention (ESTA) mechanism that assigns learnable weights to both spatial and temporal features. Specifically, two learnable attention weight matrices Att_{slow} and Att_{fast} are point-wised multiplied by the output features of the slow and fast pathways respectively. Then, the refined features are concatenated and forwarded into a fully connected layer to obtain the action recognition score for each class. Thus, the final representation R_f can be formulated as an attention summation as shown below.

$$R''_{slow} = \sum_{c \in C} R'_{slow}[c] \odot Att_{slow} \quad (10)$$

$$R''_{fast} = \sum_{c \in C} R'_{fast}[c] \odot Att_{fast} \quad (11)$$

$$R_f = Linear([\phi(R''_{slow}); \phi(R''_{fast})]) \quad (12)$$

where $\odot$ stands for the point-wise product of two matrices, $\phi(*)$ includes average pooling and batch normalization operations, $[*; *]$ denotes the concatenation operator of two tensors. The cross-entropy is eventually employed to obtain the predicted action.

With the proposed four modules mentioned above, the raw event data is effectively transformed into a better representation incorporating multiple event information, and the performance of the event-based action recognition is further improved by the design of the Slow-Fast structure and the Spatial Temporal Attention mechanism. The experimental results also confirm that all the proposed modules can have a positive effect in improving the accuracy of action recognition, as can be observed in Section V-D.

IV. THU^{E-ACT}-50 BENCHMARK

Real-world datasets for event-based action recognition include PAF [7], N-HAR [9], DHP19 [8], and the recently released DailyAction [10]. PAF [7] is a dataset recorded in the office, containing 10 action categories with 450 recordings with an average length of 5 seconds. DHP19 [8] is an event dataset focused on the task of pose estimation, thus can also be utilized for action recognition as well. It has 33 movements with 557 recordings captured in a lab using four DVS cameras. However, the action categories of DHP19 are based on body part movements, which are not practical, e.g. left arm abduction, right arm abduction, hop right foot, hop left foot, etc. DailyAction [10] is a recently released dataset for event-based action recognition,

having 1,440 recordings and 12 action categories. Since it is captured using DVS128 [44], the resolution is only 128×128 . N-HAR [9] is a category-unbalanced indoor dataset using an ATIS camera, including 5 actions, 30 subjects, and 3,091 videos. However, compared with the frame-based action recognition benchmarks, event-based datasets are too limited in size and the actions tend to be too basic. In fact, not all actions contained in the frame-based action recognition datasets are suitable for event-based recognition, since semantic details are often not visible in the event camera. Nevertheless, in some particular occasions, such as indoor health monitoring, event cameras have advantage in terms of both low power consumption and privacy encryption.

Given such considerations, we present the first real-world event-specific action recognition dataset named THU^{E-ACT}-50, along with the affiliated challenging dataset called THU^{E-ACT}-50-CHL, whose total size is more than 4 times larger than the current largest event-based action recognition dataset. For both datasets, we have deliberately selected 50 event-specific action categories that have the potential for application in practice, mainly focusing on whole-body motions, e.g. walking, running, jump up, waving hands, squatting, punch, tai chi, squat down, long jump, and indoor healthcare, e.g. falling down, stomachache, headache, vomit, back pain, staggering. Several detail-oriented actions are added, e.g. nodding, shaking head, clap, rub hands, knead face, wipe face, thumb up, check time. We have also recorded some confusing actions, e.g. standing up, sitting down, and standing after sitting. As for the human-object interaction, a number of actions are involved, e.g. calling with phone, swing objects, throw, fan, open or close umbrella, put on or take off glasses/bag, pick up, drink water. In addition to the single-player actions mentioned above, two-player interactive movements are also included, e.g. handshakes, fighting, handing over objects, and lifting chairs. Table III lists all the action categories.

For the standard THU^{E-ACT}-50, we have used the CeleX-V [45] event camera to capture action sequences from two frontal views in a room of nearly 100 square meters, with a spatial resolution of 1280×800 , which is higher than all current datasets. The two event cameras were placed in two oblique front positions of the actor, with synchronized trigger signals to ensure that they start and end shooting at the same time. The dataset contains 105 socially recruited subjects and 10,500 video recordings covering men and women of all ages (covering 15-72 years), bringing exponential scale expansion compared to the existing datasets (over 20 times for PAF [7] and DHP19 [8], over 7 times for DailyAction [10], and over 4 times for N-HAR [9]). Table II summarizes the advantages of our benchmark over existing datasets, while Fig. 3 displays some sampled sequences.

Apart from the standard dataset, we have also collected a relatively small dataset under challenging scenarios named THU^{E-ACT}-50-CHL, which is still larger than most current event-based action recognition datasets. It contains 2,330 recordings of 18 students (10 females and 8 males, aged 18-25 years old) captured from different perspectives using a DAVIS346 [46] event camera in two scenarios, i.e., a long corridor and an open

TABLE II
COMPARISONS BETWEEN THE SCHEDULED PUBLISHED BENCHMARKS (THU^{E-ACT}-50, THU^{E-ACT}-50-CHL) AND OTHER AVAILABLE DATASETS FOR EVENT-BASED ACTION RECOGNITION

Benchmark	Category Num.	Recording Num.	Subject Num.	Resolution	Sensor	Practical Actions	Event-specific Actions
PAF [7]	10	450	15	346 × 260	DAVIS 346	✓	
DHP19 [8]	33	557	17	346 × 260	DAVIS 346		
DailyAction [10]	12	1,440	15	128 × 128	DVS 128	✓	
N-HAR [9]	5	3,091	30	304 × 240	ATIS	✓	
THU ^{E-ACT} -50-CHL	50	2,330	18	346 × 260	DAVIS 346	✓	✓
THU ^{E-ACT} -50	50	10,500	105	1280 × 800	CeleX-V	✓	✓
Total	50	12,830	123	-	-	-	-

The two ready-released benchmarks contain well-designed actions that are suitable for inference using event cameras in practice.

TABLE III
LIST OF ACTIONS IN OUR CONSTRUCTED THU^{E-ACT}-50 BENCHMARK

A0: Walking	A10: Cross arms	A20: Calling with phone	A30: Fan	A40: Check time
A1: Running	A11: Salute	A21: Reading	A31: Open umbrella	A41: Drink water
A2: Jump up	A12: Squat down	A22: Tai chi	A32: Close umbrella	A42: Wipe face
A3: Running in circles	A13: Sit down	A23: Swing objects	A33: Put on glasses	A43: Long jump
A4: Falling down	A14: Stand up	A24: Throw	A34: Take off glasses	A44: Push up
A5: Waving one hand	A15: Sit and stand	A25: Staggering	A35: Pick up	A45: Sit up
A6: Waving two hands	A16: Knead face	A26: Headache	A36: Put on bag	A46: Shake hands (two-players)
A7: Clap	A17: Nod head	A27: Stomachache	A37: Take off bag	A47: Fighting (two-players)
A8: Rub hands	A18: Shake head	A28: Back pain	A38: Put object into bag	A48: Handing objects (two-players)
A9: Punch	A19: Thumb up	A29: Vomit	A39: Take object out of bag	A49: Lifting chairs (two-players)

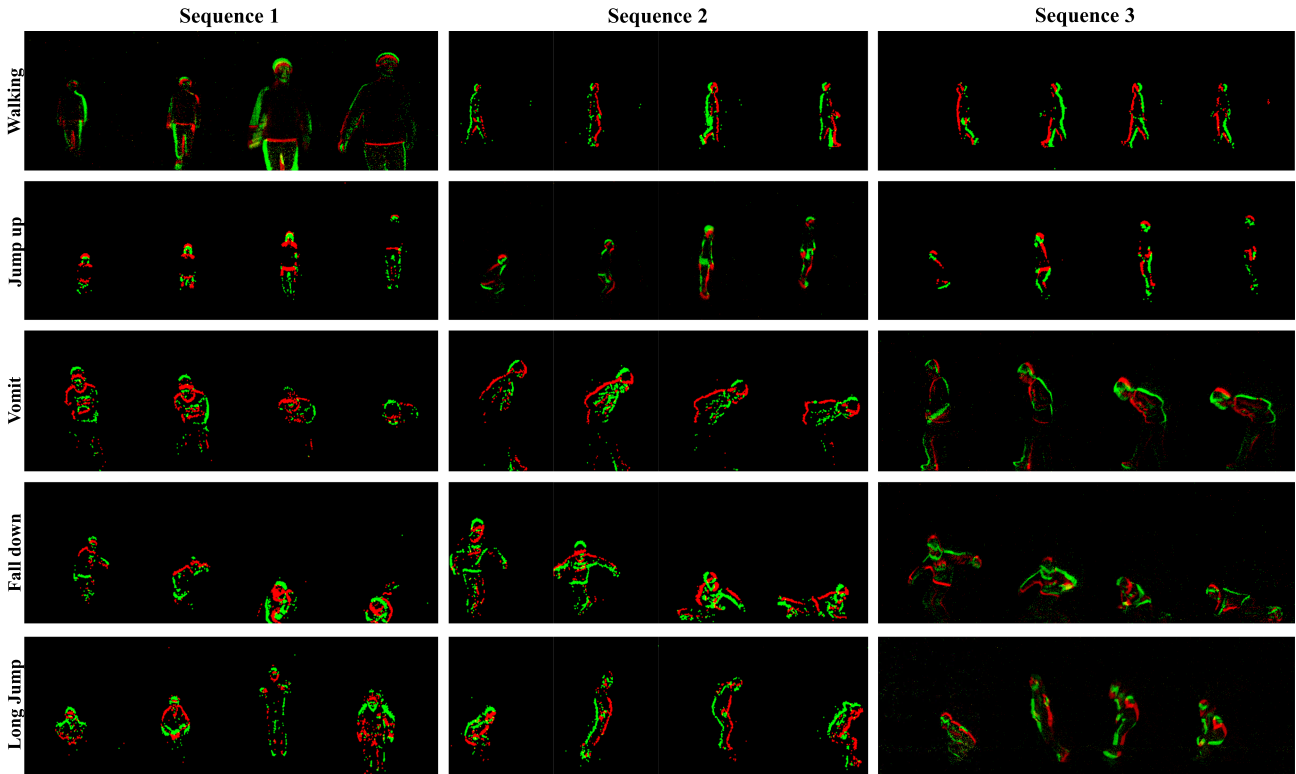


Fig. 3. Recording examples in our THU^{E-ACT}-50 dataset.

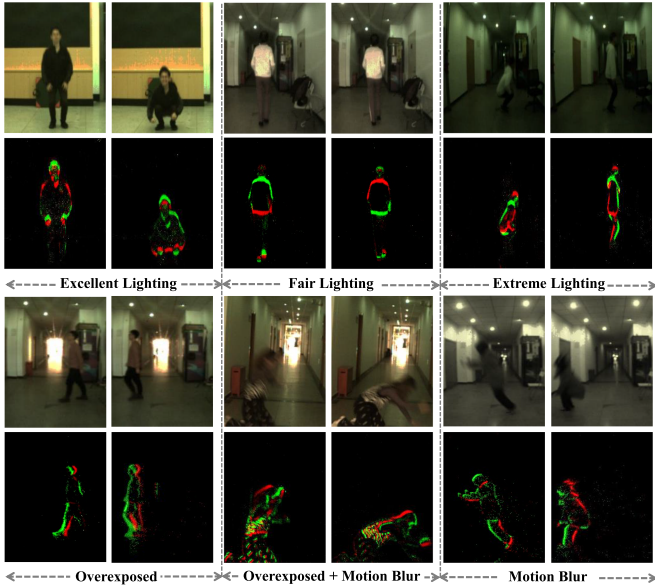


Fig. 4. Challenging conditions in the proposed $\text{THU}^{\text{E-ACT}}\text{-50-CHL}$ dataset.

hall. Each recording is within 2-5 seconds with a spatial resolution of 346×260 . Most actions are recorded from the front, left, and right views, and some from the back view. Compared to the standard $\text{THU}^{\text{E-ACT}}\text{-50}$, the $\text{THU}^{\text{E-ACT}}\text{-50-CHL}$ is more challenging in terms of illumination conditions (as shown in Fig. 4), which leads to more noisy event stream, and smaller and less detectable action magnitude. Therefore, it serves as a rigorous benchmark to evaluate action recognition methods.

V. EXPERIMENTS AND RESULTS

A. Evaluation Protocols

Since neither PAF [7], DHP19 [8] nor DailyAction [10] provides any evaluation protocol for action recognition task except for the dataset partitioning in DHP19, we split the training and test set in a ratio of 8:2 for PAF and DailyAction, and used the split provided by [8] for DHP19. As DHP19 is captured from four different perspectives, the difficulty of action recognition from different perspectives is incomparable, e.g. the motion details under the back view may not be captured. Hence, we trained and tested separately under each camera view, denoted as DHP-C1 to DHP-C4. As for the proposed $\text{THU}^{\text{E-ACT}}\text{-50}$ and $\text{THU}^{\text{E-ACT}}\text{-50-CHL}$ datasets, we also divided the subjects in a ratio of 8:2 to make the identity sets in the constructed training and test sets disjoint. The $\text{THU}^{\text{E-ACT}}\text{-50}$ training and test sets contain 85 and 20 persons respectively, while the $\text{THU}^{\text{E-ACT}}\text{-50-CHL}$ training and test sets contain 14 and 4 persons respectively. In DHP19, $\text{THU}^{\text{E-ACT}}\text{-50}$ and $\text{THU}^{\text{E-ACT}}\text{-50-CHL}$ datasets, the Top-1, Top-3 and Top-5 accuracy are reported as evaluation metrics. In the PAF dataset, the Top-1, Top-2, and Top-3 are reported because of their category size. Only Top-1 is reported in DailyAction, as subsequent Top-N metrics have all reached 100% under our proposed framework.

B. Implementation Details

The proposed method in this paper is comprised of four modules. To demonstrate the effectiveness of each proposed module, we also design a baseline method that converts the raw event data into a frame-like representation in a simple and handcrafted way and directly feeds it into the backbone network instead of adopting the EVF, LMF, ESNF, and ESTA. In our experiments, ResNet 34 [48] pre-trained on ImageNet [49] is employed as the backbone network to meet the low-power and low-latency requirements of event processing. The hyperparameter α in (4) for adjusting the timestamp scale is chosen as 300, the number of time bins for the slow and fast pathways is set to $T_{\text{slow}} = 4$ and $T_{\text{fast}} = 16$, respectively. The network is trained for 80 epochs on all three benchmarks using the Adam [50] optimizer with a learning rate of 1×10^{-4} and a weight decay of 1×10^{-4} . Exponential learning rate decay [51] strategy is applied, with the gamma of 0.5. All experiments are implemented based on PyTorch [52], and training with a Tesla V100 GPU.

C. Quantitative Results

1) *Comparisons With State-of-the-Arts*: We have conducted experiments on PAF, DailyAction, $\text{THU}^{\text{E-ACT}}\text{-50-CHL}$ and $\text{THU}^{\text{E-ACT}}\text{-50}$ datasets (shown in Table IV) to demonstrate the superiority of our proposed method over existing event-based action recognition works, i.e., HMAX SNN [47] and Motion SNN [10]. Our framework achieves a significant improvement in Top-1 accuracy, outperforming the state-of-the-art (SOTA) with a 14.5%, 7.6%, 11.2%, and 7.4% increase on PAF, DailyAction, $\text{THU}^{\text{E-ACT}}\text{-50-CHL}$, and $\text{THU}^{\text{E-ACT}}\text{-50}$, respectively. For $\text{THU}^{\text{E-ACT}}\text{-50-CHL}$ and $\text{THU}^{\text{E-ACT}}\text{-50}$, the results of HMAX SNN and Motion SNN are based on code replication.

It can be inferred that one of the reasons that the proposed framework achieves a substantial improvement lies in the CNN backbone. Previous SNN-based works are difficult to train due to limited effective back-propagation algorithms, hindering their performance on event-based action recognition. By converting raw event data into frame-like representations, the proposed method takes full advantage of CNNs, which have stronger generalization capabilities and easier training difficulties than SNNs. Therefore, even adopting the most general CNN backbone network, e.g., ResNet, for action classification can achieve better performance than well-designed SNN-based approaches. Another aspect of performance boost comes from the four proposed modules (including LMF, EVF in the front end, and ESNF, ESTA in the back end), each of which can further boost the performance of event-based action recognition to varying degrees, as detailed below.

2) *Comparisons With Baseline*: Table V shows the quantitative results of the module combinations on the PAF, DHP19, $\text{THU}^{\text{E-ACT}}\text{-50}$ and $\text{THU}^{\text{E-ACT}}\text{-50-CHL}$ datasets. Since the action in the 4th camera view is clearest in the DHP19 dataset, we use only DHP19-C4 for our main experiments, while the results of our full model on DHP19-C1 to DHP19-C3 are reported in Table VI. It can be seen that the proposed method outperforms the baseline (B/L) on small datasets PAF and DHP19. On PAF, the Top-1, Top-2, and Top-3 accuracy improve by 6.5%, 4.4%,

TABLE IV
COMPARISON OF TOP-1 ACCURACY BETWEEN THE PROPOSED METHOD AND THE STATE-OF-THE-ARTS ON FOUR BENCHMARKS

Method	PAF [7]	DailyAction [10]	THU ^{E-ACT} -50-CHL	THU ^{E-ACT} -50
HMAX SNN [47]	55.0	68.3	32.7	65.1
Motion SNN [10]	78.1	90.3	47.3	85.3
EV-ACT (Ours)	92.6	97.9	58.5	92.7

TABLE V
COMPARISON OF RESULTS USING INCREMENTAL PROPOSED MODULES ON THREE BENCHMARKS, INCLUDING EVENT VOXEL FILTERING (EVF), LEARNABLE MULTI-REPRESENTATION FUSION (LMF), EVENT-BASED SLOW-FAST NETWORK (ESFN), AND EVENT SPATIAL TEMPORAL ATTENTION (ESTA) MECHANISMS

Modules					PAF [7]			DHP19-C4 [8]			THU ^{E-ACT} -50-CHL			THU ^{E-ACT} -50		
B/L	LMF	EVF	ESFN	ESTA	Top-1	Top-2	Top-3	Top-1	Top-3	Top-5	Top-1	Top-3	Top-5	Top-1	Top-3	Top-5
✓					86.1	93.8	97.2	86.6	95.8	98.1	50.7	74.2	82.1	85.9	95.5	97.0
✓	✓				88.3	94.4	96.3	88.5	97.2	98.5	53.2	75.0	83.0	89.4	96.4	98.1
✓	✓	✓			88.5	94.4	98.2	89.0	98.8	99.4	53.8	75.5	83.0	90.0	96.8	98.5
✓	✓	✓	✓		88.9	94.4	94.4	89.7	97.6	99.4	55.9	75.7	83.2	91.2	97.5	98.6
✓	✓	✓		✓	90.7	96.3	98.2	90.3	98.2	99.4	55.4	76.2	84.9	91.5	97.8	99.0
✓	✓		✓	✓	91.5	96.3	100	89.8	98.2	98.5	57.5	77.7	86.0	92.3	98.1	99.2
✓	✓	✓	✓	✓	92.6	98.2	100	91.5	98.8	99.4	58.5	78.6	86.0	92.7	98.4	99.2

TABLE VI
ACTION RECOGNITION ACCURACY OF OUR FULL MODEL ON THE DHP19-C1, DHP19-C2, AND DHP19-C3 SUBSETS

Camera View	Top-1	Top-3	Top-5
DHP19-C1	78.0	93.9	98.8
DHP19-C2	82.9	95.7	96.3
DHP19-C3	84.2	93.3	97.6

and 2.8%, respectively. On DHP19, the Top-1, Top-3, and Top-5 accuracy improve by 4.9%, 3.0%, and 1.3%, respectively. As for our large-scale datasets THU^{E-ACT}-50 and THU^{E-ACT}-50-CHL, the proposed method achieves significant improvements, with 7.8%, 4.4%, 3.9% on the Top-1, Top-3, Top-5 accuracy for THU^{E-ACT}-50-CHL, and 6.8%, 2.9%, 2.2% for the standard THU^{E-ACT}-50.

Moreover, Table V also proves that all of the proposed modules play indispensable roles in action recognition. The LMF module, which incorporates multiple event representations in a learnable manner, outperforms using a single representation. The EVF module reduces memory and power load through down-sampling while denoising improves network performance, indicating the existence of redundancy in the raw event data. The ESFN dual-branch architecture enhances appearance and motion features separately with different temporal granularity. The performance boost obtained by ESFN on THU^{E-ACT}-50-CHL is higher than on PAF and DHP19, due to the inclusion of complex actions in challenging environments requiring inference based on both slow-based semantic and fast-based motion features. The proposed ESTA is also effective for event data, attending to motion-dense regions with a high number of events across spatio-temporal domains.

3) *Comparisons With Frame-Based Pipeline:* Since the captured THU^{E-ACT}-50-CHL dataset has RGB frame-based sequences in addition to event data, we conducted experiments on the THU^{E-ACT}-50-CHL under different input sources. Specifically, we have considered four different input sources: pure RGB sequences, frame differentiation, optical flow, and the proposed event representation LMFR. The frame differentiation and optical flow are computed with respect to time from pure RGB sequences. The back-end structure of the proposed framework, which includes the ESFN and ESTA modules, is used for training and testing. Table VIII shows the Top-1 accuracy achieved for each input source.

The extreme variability of illumination in THU^{E-ACT}-50-CHL makes it difficult for the network to filter out invariant backgrounds and noise, resulting in only 30.8% accuracy when using pure RGB as input, and only 12.0% accuracy without the proposed ESTA module. However, using frame differentiation or optical flow calculated from RGB sequences can partially remove background information, resulting in notable performance gains. Specifically, with the ESTA module, frame differentiation achieves 44.4% accuracy (20.6% without), and optical flow achieves 46.8% accuracy (21.3% without). By contrast, the event representation exemplified by the proposed LMFR exhibits significantly better performance, with an accuracy of 58.5% (55.9% without) using the same network structure and number of parameters. This indicates that event cameras, which only record motion without invariant information, can be particularly effective in challenging illumination environments where traditional frame-based cameras may struggle to differentiate between foreground and background elements or motion and stillness. Thus, event-based action recognition shows significant superiority over traditional frame-based approaches in these scenarios.

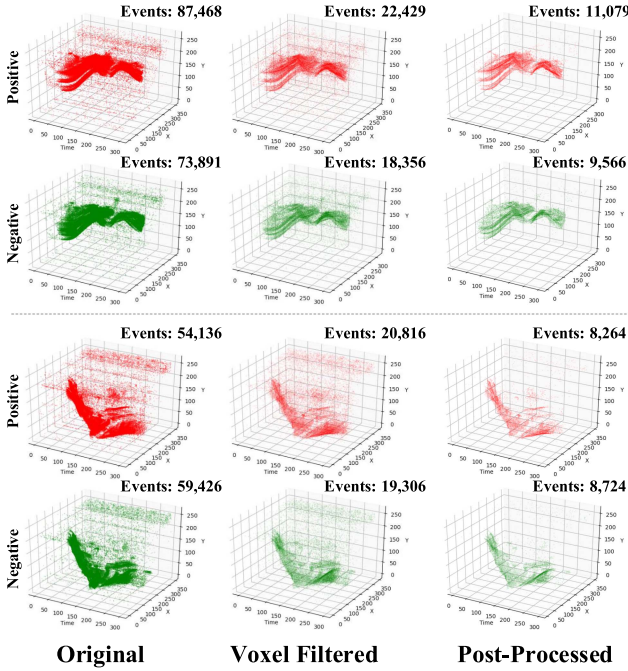


Fig. 5. Visualization of two sampled event streams under the EVF module in THU^{E-ACT}-50, with red for positive events and green for negative events.

D. Ablation Study

We demonstrate the effectiveness of each proposed module by incrementally adding them to the baseline method and visualizing intermediate information as well. Table V presents the results of the quantitative experiments.

Event Voxel Filtering: Table V indicates that removing the EVF module leads to a Top-1 accuracy decrease of 1.1%, 1.7%, 1.0%, and 0.4% on PAF, DHP19, THU^{E-ACT}-50-CHL, and THU^{E-ACT}-50 datasets, respectively. To further demonstrate the effectiveness of the EVF module, we visualize the down-sampling and denoising results of two event streams from our THU^{E-ACT}-50 dataset in Fig. 5, with red and green points representing positive and negative events. The first column shows the original event stream, the second column displays the event data after voxel filtering, and the third column visualizes the result after applying our proposed variance-based post-processing. It can be observed that the original event stream generates more than 100,000 events, including positive and negative polarity, with a considerable amount of background noise. Voxel filtering reduces the number of event points by a large margin, but still leaves much noise. However, our variance-based post-processing effectively removes almost all the background noise, decreasing the number of events by an average of 5-10 times through the entire EVF module. This significant reduction in the burden of event memory and processing while retaining prominent spatio-temporal features highlights the usefulness of the EVF module.

Moreover, we investigate the denoising effect of different spatio-temporal voxel sizes and compare it with the random down-sampling method, as shown in Table VII. Our experimental setup employs a set of two-by-two combinations of

spatial voxel sizes (S-voxel size) and temporal voxel sizes (T-voxel size), ranging from 1 to 10, with the number of denoised event points (Points), Chamfer Distance (CD) [53], Hausdorff Distance (HD) [54], and average processing time (Time) as evaluation metrics. The CD measures the average distance between each denoised event point and its nearest point in the original point set, while the HD measures the distance between the farthest outlier and its corresponding point in the original point set. The experiments are conducted under identical inputs and settings, with an average input of 59,426 event points. As spatial and temporal voxel size increase, the number of denoised event points and processing time decrease. It can be observed that the proposed EVF module outperforms the random down-sampling strategy in all combinations with more stability. Regarding denoising metrics, the CD is inversely related to the number of retained event points. However, a smaller CD value only indicates the smaller gap between the original and the denoised event sets, which does not fully reflect the better filtering effect in this task, so we use HD as the main reference standard.

It can be inferred that the sizes of spatial-temporal voxels should be chosen within a moderate range. Too small voxel size retains redundant event points, increases denoising time, and burdens the subsequent network inference. On the other hand, too large a size causes information loss despite faster processing speed. The proposed method achieves the lowest HD value of 57.54 when S-voxel size = 5×5 and T-voxel size = 5, while a scene with S-voxel size = 1×1 has a worse HD value. Furthermore, the S-voxel size and the T-voxel size should not differ too much, as long striped voxel cubes are created, which are not suitable for integrating information. Therefore, we choose the combination of S-voxel size = 5×5 and T-voxel size = 5 in the main experiments. The average time consumed by the EVF module, including the proposed post-processing mechanism, is 0.110 seconds, which is only about 6% more time-consuming than the 0.103 seconds of the original voxel filtering.

Event Intermediate Representations: Since there is no clear consensus on the effectiveness of various event representations, we have compared several mainstream event representations, including Event Frame [11], Event Count [14], [30], Event Voxel [15], Surface of Active Events (SAE) [12], Event Spike Tensor (EST) [35], and our proposed Learnable Multi-Fusion Representation (LMFR), on the THU^{E-ACT}-50-CHL and THU^{E-ACT}-50 datasets presented in this paper. The Top-1 accuracy is shown in Fig. 8. Among them, Event Frame records polarity, Event Count and Event Voxel record count, while Surface of Active Events (SAE) and Event Spike Tensor (EST) record timestamp information. Regarding single representations, Event voxel and EST exhibit the highest accuracy on both datasets, while Event Frame performs the worst. Notably, our proposed Learnable Multi-Fusion Representation (LMFR) achieves a significant accuracy improvement on both datasets by accommodating information from multiple event representations.

Learnable Multi-Fused Representation: As shown in Table V, the proposed Learnable Multi-representation Fusion (LMF) can remarkably improve the performance of action recognition.

TABLE VII
COMPARISON OF THE PROPOSED EVENT Voxel FILTERING MODULE WITH RANDOM DOWN-SAMPLING AT DIFFERENT SPATIAL Voxel SIZE (S-Voxel SIZE) AND TEMPORAL Voxel SIZE (T-Voxel SIZE)

S-voxel size	Method	T-voxel size=1				T-voxel size=3				T-voxel size=5				T-voxel size=10			
		Points	CD	HD	Time	Points	CD	HD	Time	Points	CD	HD	Time	Points	CD	HD	Time
1×1	Random	14,270	4.85	52.27	-	12,156	5.07	57.82	-	9,895	5.30	59.22	-	8,901	5.51	69.14	-
	Ours	14,270	3.48	68.98	0.41	12,156	3.60	61.44	0.39	9,895	3.72	63.87	0.36	8,901	3.81	66.87	0.39
3×3	Random	9,403	5.30	70.31	-	8,264	5.70	65.91	-	7,420	5.80	62.98	-	6,284	5.95	67.70	-
	Ours	9,403	3.68	68.98	0.29	8,264	3.78	61.11	0.21	7,420	3.93	61.26	0.17	6,284	4.17	66.32	0.14
5×5	Random	8,487	5.58	63.43	-	6,236	6.10	74.51	-	5,343	6.31	97.10	-	4,340	7.28	78.39	-
	Ours	8,487	3.78	58.02	0.22	6,236	4.02	58.02	0.13	5,343	4.22	57.54	0.11	4,340	4.49	57.56	0.09
10×10	Random	6,336	5.98	71.05	-	4,241	6.82	108.66	-	3,630	7.05	90.32	-	2,839	7.30	88.00	-
	Ours	6,336	4.23	59.04	0.13	4,241	4.63	57.84	0.09	3,630	5.10	68.85	0.09	2,839	5.58	61.78	0.05

The average number of input event points is 59,426, and the evaluation metrics include the number of denoised event points (Points), chamfer distance (CD), hausdorff distance (HD), and average processing time (Time).

TABLE VIII
RESULTS OF THE PROPOSED FRAMEWORK ON TOP-1 ACCURACY WITH DIFFERENT INPUT SOURCES (PURE RGB SEQUENCE, IMAGE DIFFERENTIATION, OPTICAL FLOW, AND THE PROPOSED EVENT REPRESENTATION LMFR) IN THE CHALLENGING THU^{E-ACT}-50-CHL DATASET

Input	w/o Att.	with Att.
Pure RGB	12.0	30.8
Frame Differentiation	20.6	44.4
Optical Flow	21.3	46.8
Event Repr. (LMFR)	55.9	58.5

TABLE IX
IMPACT OF THE PROPOSED LEARNABLE MULTI-FUSED REPRESENTATION

Repr.	THU ^{E-ACT} -50-CHL			THU ^{E-ACT} -50		
	Top-1	Top-3	Top-5	Top-1	Top-3	Top-5
Event Count	54.3	75.7	84.3	89.8	97.2	98.3
Event Frame	56.0	77.5	84.3	90.2	96.8	98.3
Time Surface	55.4	76.2	84.9	89.4	97.0	98.0
Multi-Fused	58.1	77.9	84.9	91.9	97.8	99.2
Multi-Fused + Learn Repr.	58.5	78.6	86.0	92.7	98.4	99.2

Compared to the baseline, the Top-1 accuracy improves 2.2%, 1.9%, 2.5%, 3.5% on PAF, DHP19, THU^{E-ACT}-50-CHL, and THU^{E-ACT}-50. To further verify the advantages of the proposed Learnable Multi-Fused Representation, we have conducted experiments under three scenarios: single event representation (Event Count, Event Frame, Time Surface), multi-representation fusion, and learnable multi-representation fusion, with the results shown in Table IX. The Multi-Fused Representation, obtained by concatenating these three single event representations after normalization, exhibits significant improvement, i.e., 3.8%, 2.1%, and 2.7% on THU^{E-ACT}-50-CHL and 2.1%, 1.7%, and 2.5% Top-1 improvement on THU^{E-ACT}-50 compared to Event Count, Event Frame, and Time Surface, respectively. It can be reasoned that the fusion of multiple event representations can be seen as a feature enhancement that allows the network to obtain

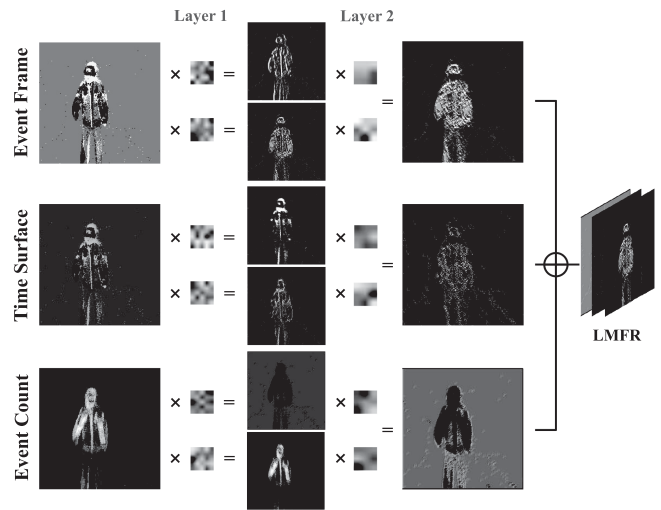


Fig. 6. Visualization of filters and feature maps in the proposed Learnable Multi-Fused Representation (LMFR).

diverse information including event polarity, counts, and time-stamps simultaneously rather than a single feature. Furthermore, the learnable feature transformation with a convolutional layer can further improve the performance by gaining 0.4% and 0.8% Top-1 improvement on THU^{E-ACT}-50-CHL and THU^{E-ACT}-50 respectively.

To better understand the role of the proposed Learnable Multi-fused Representation (LMFR), we visualize the filters and feature maps within it, as depicted in Fig. 6. Each single event representation is separately transformed by 2-layer convolutional kernels, and the resulting feature maps are concatenated to form the LMFR. The Event Frame, Time Surface, and Event Count representations of a sample within a given time window are shown in the first column. It can be observed that the three single representations vary significantly, and the Event Frame and Time Surface possess obvious positive and negative polarity differences. The first convolutional layer is responsible for extracting features for the two distinct polarities and obtaining semantic features for the two polarities separately. In the second convolutional layer, the two channels of each filter

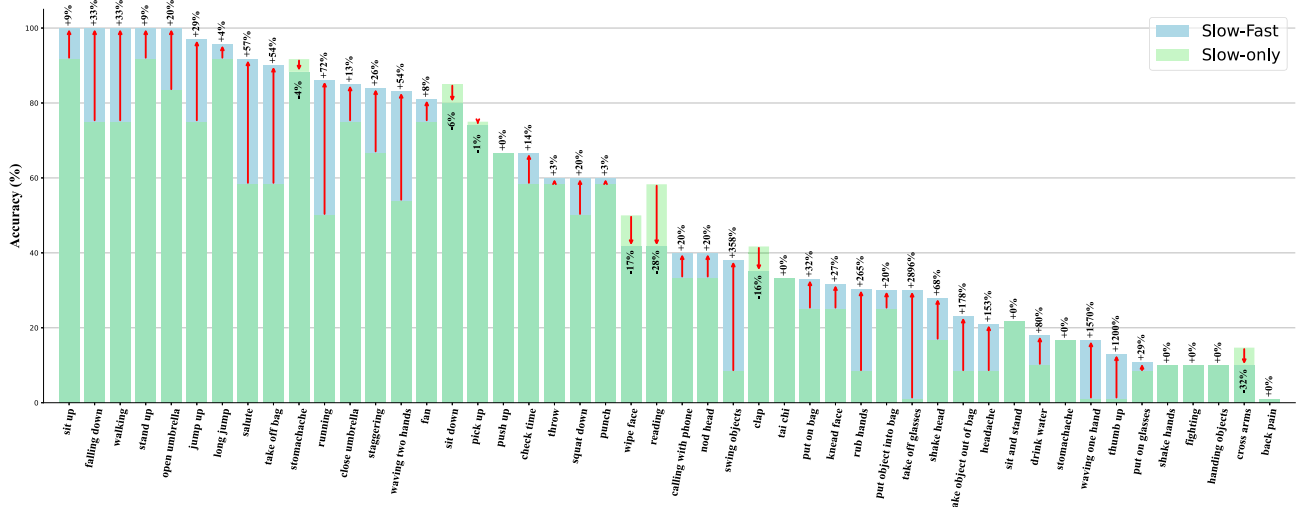


Fig. 7. Accuracy comparison of the Event-based Slow-Fast network and Slow-only structure on each action category under the THU^{E-ACT}-50-CHL.

TABLE X
IMPACT OF THE COMBINATIONS OF SLOW-FAST TIME BINS FOR THE
EVENT-BASED SLOW-FAST NETWORK (ESFN)

Num. of Bin		DHP19-C4 [8]			THU ^{E-ACT} -50-CHL		
T_{slow}	T_{fast}	Top-1	Top-3	Top-5	Top-1	Top-3	Top-5
2	4	90.2	95.7	98.2	56.0	75.8	83.2
2	8	90.9	97.0	98.8	57.5	77.7	86.0
2	16	91.5	97.0	98.8	57.6	77.1	83.7
4	8	91.0	98.2	98.8	58.1	78.6	85.7
4	16	91.2	97.6	99.4	58.5	78.6	86.0
8	16	91.5	98.8	99.4	58.0	78.6	86.6

demonstrate nearly opposite color distributions, especially in the Event Frame and Time Surface. This observation suggests that the second convolutional layer is responsible for fusing the features extracted under different polarities and unifying the three representations into a more discriminative pattern.

Number of Time Bins: We have conducted a sensitivity analysis on the number of time bins T for both single-branch and dual-branch within the Event-based Slow-Fast Network (ESFN). Fig. 10 illustrates the effect of different T on the Top-1 accuracy under a single-branch network, excluding the slow-fast pathways. It can be observed that the framework performs poorly when T is particularly small, e.g., set to 2, since a small number of the time bin quantifies events in a relatively long temporal window into one dimension, causing the intermediate representation to lose too much information. The performance is insensitive to T in the range from 4 to 18, where slightly better results are obtained in the intervals $T \in [6, 8]$ and $T \in [16, 18]$. A value of $T > 18$ would create an excessive memory load unsuitable for practical use. This suggests that $T \in [6, 8]$ better accrues the spatial semantics of appearance, while $T \in [16, 18]$ can capture motion dynamics over short time periods, further confirming the necessity of designing the slow and fast dual-pathways.

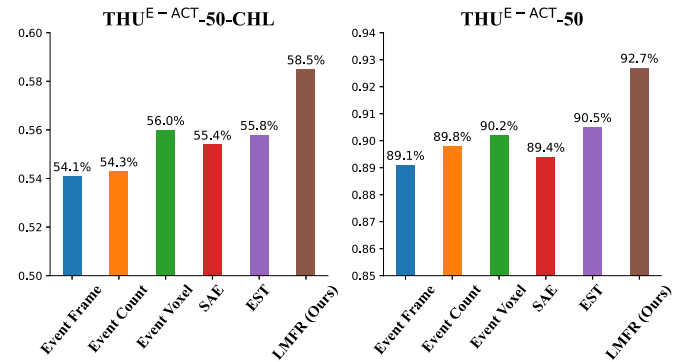


Fig. 8. Comparison of top-1 accuracy of various event representations on THU^{E-ACT}-50-CHL and THU^{E-ACT}-50 datasets.

Event-Based Slow-Fast Network: We have also conducted a sensitivity analysis for the ESFN structure by varying the number of time bins T_{slow} and T_{fast} in the slow and fast pathways, respectively. The results are presented in Table X. Our analysis shows that the effect of different combinations of T_{slow} and T_{fast} is not significant, as long as they are within appropriate ranges. Moreover, larger variations of T_{slow} and T_{fast} lead to better performance due to the diverse event features extracted by the two branches.

To further demonstrate the effectiveness of the proposed ESFN structure, we have compared the network structure containing both slow and fast dual branches with one containing only the slow branch. The Slow-Fast network uses $T_{slow} = 6$ and $T_{fast} = 18$ time bins, while the Slow-only network uses $T_{slow} = 6$ bins. We evaluate the accuracy of all action categories under the THU^{E-ACT}-50-CHL dataset, as depicted in Fig. 7. It can be observed that the Slow-Fast network significantly improves the accuracy of 32 action categories and decreases the accuracy of only 7. The accuracy for high-precision actions

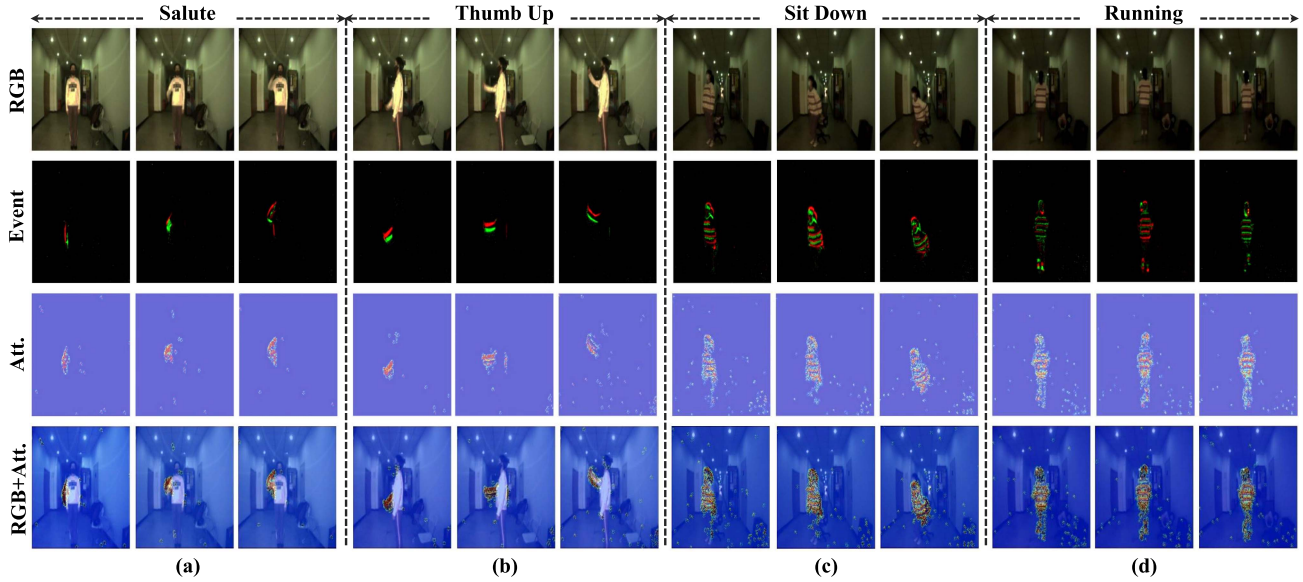


Fig. 9. Visualization of attention maps in the proposed ESTA module on the $\text{THU}^{\text{E-ACT-50-CHL}}$ dataset. Among them, samples (a) and (b) are movements of localized body parts, while (c) and (d) display movements in which the entire body is involved.

such as “sit up”, “falling down”, “walking”, “stand up”, and “open umbrella”, is improved to 100%. In terms of relative improvement, “take off glasses,” “waving one hand,” “thumb up,” and “swing objects” improves the most. These significantly improved actions are all temporally informative ones that are difficult to infer from static representations brought by the Slow-only structure, while the Slow-Fast boosts their capture of rapid movement changes by fusing the fast branch. For a few actions with little temporal variation, such as “wipe face”, “reading”, “clap”, “cross arms”, excessive temporal segmentation results in a very limited number of event points in each time bin, resulting in decreased accuracy. Overall, the proposed Slow-Fast structure achieves significant improvement in most action categories, while the degradation of actions is relatively weak *versus* others’ increase.

Event-based Spatial Temporal Attention: Table V shows that removing the ESTA module from the dual pathways results in a decline in Top-1 accuracy of 3.7%, 1.8%, 2.6%, and 1.5% on the PAF, DHP19, $\text{THU}^{\text{E-ACT-50-CHL}}$, and $\text{THU}^{\text{E-ACT-50}}$ datasets, respectively. In contrast, the inclusion of the ESTA module in the single-branch structure improves Top-1 accuracy by 2.2%, 1.3%, 1.6%, and 1.5%, respectively. This suggests that the ESTA module can locate the occurrence time and location of the action effectively and assign higher attention weights to those critical areas. To further confirm its effectiveness, we have visualized attention maps on the $\text{THU}^{\text{E-ACT-50-CHL}}$ dataset, as shown in Fig. 9. The first and second rows show the RGB image and event sequence for the corresponding sample. The third row displays the visualized pure attention map, and the fourth row shows the result of adding the attention map to the RGB image. The selected samples include two local actions (Salute and Thumb up) and two full-body movements (Sit down and Running). As can be observed, the ESTA module successfully locates the moving areas and assigns larger weights to regions

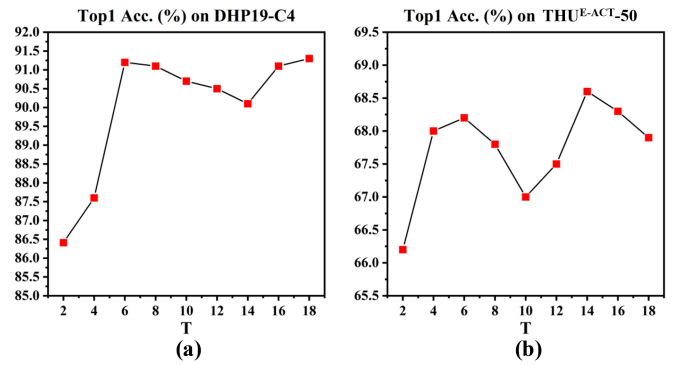


Fig. 10. Impact of the number of time bins under single-branch on the DHP19-C4 and $\text{THU}^{\text{E-ACT-50-CHL}}$.

with motion for local actions, while for full-body movements, it extracts the entire body area as all parts of the body are moving. Compared to RGB image sequences where it is difficult to distinguish foreground and background, the characteristics of event data make it easier for attention to focus on the moving foreground area, demonstrating the superiority of event-based action recognition methods over frame-based ones.

E. Cross-View Evaluation

Since the $\text{THU}^{\text{E-ACT-50}}$ dataset presented in this paper contains two distinct views (referred to as View A and View B), we have performed experiments across views and compared it with previous event-based action recognition methods, including HMAX SNN [47] and Motion SNN [10]. Specifically, we have conducted under four settings: 1) training and testing on View A, 2) training and testing on View B, 3) training on View A and testing on View B, and 4) training on View B and testing

TABLE XI
COMPARISON OF CROSS-VIEW ACTION RECOGNITION ON THE THU^{E-ACT}-50 DATASET

Method	Train on View A		Train on View B	
	View A	View B	View B	View A
HMAX SNN [47]	66.5	25.5 (-41.0)	63.9	27.8 (-36.1)
Motion SNN [10]	87.0	57.0 (-30.0)	84.2	54.4 (-29.8)
EV-ACT (Ours)	93.8	68.3 (-25.5)	89.8	68.4 (-21.4)

on View A. The results presented in Table XI indicate performance degradation for all methods in the cross-view evaluation. However, when compared to the 41.0% and 36.1% accuracy decline observed in HMAX SNN and the 30.0% and 29.8% decline observed in Motion SNN, the proposed method exhibits smaller declines of 25.5% and 21.4% under the two settings. We believe that event representations and CNN-based networks have stronger generalization abilities than the SNN-based ones. Furthermore, the THU^{E-ACT}-50 dataset we constructed can serve as a benchmark for future research on cross-view action recognition.

F. Energy Consumption and Embedded System

To verify the efficiency of the proposed EV-ACT framework, we have analyzed its energy consumption and implemented an embedded system. Actually, energy consumption involves both sensor capture and computation. For sensor capture, Table XII compares the energy consumption of some current mainstream event cameras and frame-based cameras in terms of latency and power consumption. Event cameras typically have a latency of a few microseconds (μ s) and consume a few milliwatts (mW) of power, yet frame-based cameras have a latency of a few milliseconds (ms) and consume a few watts (W) of power, which are not in the same order of magnitude.

As for the computation of energy consumption, we deploy our action recognition system on the Nvidia Jetson AGX Xavier, a powerful single-board computer designed for edge computing. Our system consists of an event camera DAVIS346 and a monitor, connected to the Nvidia Jetson AGX Xavier, as illustrated in Fig. 11. The event data, recorded at 10 ms intervals using the official DV-GUI software provided by IniVation, is then packaged and sent to the processor. We accumulate the number of events within a time window of 5 units in length (about 50 ms) for real-time display and 300 units in length (about 3 s) for action recognition. The events within this 300-unit-length time window are accumulated and sent to the processor only when the number of triggered event points reaches the preset threshold of 5,000. Each batch of event data is regularized and then fed into the pre-trained event-based action recognition network. An action category is reported if the prediction score for that category exceeds 0.85. The top part of Fig. 11 shows a realistic “falling down” scenario under an indoor scene, while the bottom part shows the event stream and the recognition result.

To quantitatively evaluate the efficiency of the embedded system, we have experimented with its energy consumption on the test set of THU^{E-ACT}-50-CHL. We use model parameters

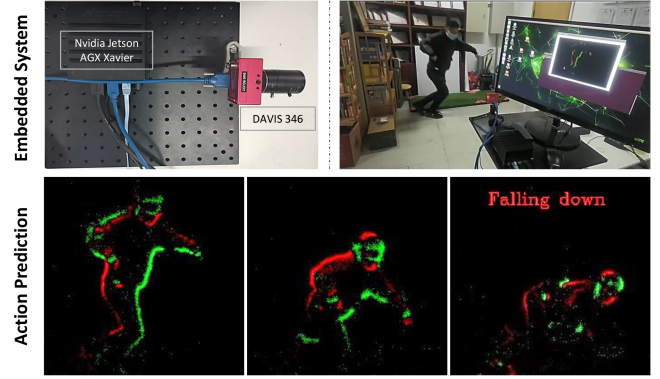


Fig. 11. Implementation of our embedded system. The DAVIS346 event camera is connected to an edge computing processor, i.e., Nvidia AGX Xavier, for real-time low-power action recognition.

(M), floating-point operations per second (GFlops), power consumption (W), and inference time per sample (ms) as evaluation metrics, where the inference time per sample is the average processing time for a single event data or image sequence input corresponding to a recording in the test set. To compare the energy consumption differences between event-based and frame-based action recognition methods, four classic frame-based action recognition methods, including TSM [25], STM [57], UniFormer [58] and I3D [37] are selected. For frame-based action recognition methods, 8 frames are sampled as input for each time, and the inference time is calculated by averaging the total processing time for all frames in a video recording. As for the proposed event-based action recognition framework, ResNet 18 and ResNet 34 are used for comparison since they can achieve high accuracy with ease, while larger models such as ResNet 50 may cause severe overfitting. The results shown in Table XIII indicate that the proposed framework only requires 11.2 M parameters, 14.5 GFlops, 5.4 W power consumption, and an average inference time of 72.8 ms per sample under ResNet 18, and 21.3 M parameters, 29.0 GFlops, 7.2 W power consumption, and an average inference time of 76.9 ms per sample under ResNet 34, all of which are much lower than those of frame-based action recognition methods. The reason is that event data only contains motion modality and not background, so ResNet 34 or ResNet 18 can easily achieve excellent performance. In addition, since event cameras almost do not trigger event points when no motion exists, the number of event points in the time window can be summed to determine whether the input needs to be fed into the network. Therefore, the energy consumption in actual applications is much lower than that shown in Table XIII.

Similar to the frame rate (Frame Per Second, FPS) in frame-based methods, we introduce the event rate (Event Per Second, EPS) to measure the number of event points that can be processed per second. The embedded event-based action recognition system deployed in this paper achieves real-time inference on Xavier, with an average of 326,140 EPS and 1.3 GB memory usage, demonstrating its ability to perform accurate and real-time action recognition with low power consumption.

TABLE XII
COMPARISON OF EVENT CAMERAS AND FRAME-BASED CAMERAS IN TERMS OF POWER CONSUMPTION FOR SENSOR CAPTURE

Type	Model	Resolution	Latency (μ s)	Power Consumption (mW)	Frame Rate (FPS)
Frame-based Cameras	PlayStation Eye	640 $\times$ 480	75,000	2,500	60
	Microsoft LifeCam Studio	1080 $\times$ 720	60,000	2,600	30
	Logitech C920	1920 $\times$ 1080	50,000	5,000	30
Event Cameras	DAVIS 240 [55]	240 $\times$ 180	12	5-14	N/A
	DAVIS 346 [46]	346 $\times$ 260	20	10-170	N/A
	ATIS [56]	304 $\times$ 240	3	50-175	N/A
	CeleX-V [45]	1280 $\times$ 800	8	400	N/A

TABLE XIII
COMPARISON OF EVENT-BASED AND FRAME-BASED ACTION RECOGNITION IN TERMS OF POWER CONSUMPTION FOR INFERENCE ON XAVIER, WHERE RN REFERS TO RESNET

Type	Method	Backbone	Params (M)	Flops (G)	Power Consumption (W)	Time Per Sample (ms)
Frame-based Action Recognition	TSM [25]	2D-RN 50	24.3	33.1	10.8	90.5
	STM [57]	2D-RN 50	24.0	33.3	11.2	95.2
	UniFormer [58]	Transformer	49.7	96.7	26.9	145.4
	I3D [37]	3D-RN 50	35.3	168.0	38.8	284.1
Event-based Action Recognition	EV-ACT (Ours)	2D-RN 18	11.2	14.5	5.4	72.8
		2D-RN 34	21.3	29.0	7.2	76.9

The event-based action recognition is expected to achieve more significant energy savings in practice than presented in the table, since the event data does not require to be fed into the network when only a few events exist.

VI. CONCLUSION

In this paper, we propose an event-based action recognition framework **EV-ACT**, which can perform low-power action recognition without directly exposing fine-grained visual details. Our proposed framework consists of an Event Voxel Filtering module, a Learnable Multi-representation Fusion module, an Event-based Slow-Fast Network, and an Event-based Spatial Temporal Attention mechanism, and achieves state-of-the-art performance. We also construct and release the two largest benchmark datasets for real-world event-specific action recognition, i.e., **THU^{E-ACT}-50** and **THU^{E-ACT}-50-CHL**, which could facilitate the development of the community. Our method has also been deployed in a mobile platform, demonstrating its potential for low-power action recognition in practice. In further work, it is necessary to investigate the multi-view event data on action representation, since the actions can happen in any view direction. It is also crucial to explore more suitable network architectures that can enable efficient event-based action recognition.

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